

text: a b c d e f g h i j k

pattern: c d e f

$$h(cdef) = h(3, 4, 5, 6) = 18$$

a	1
b	2
c	3
d	4
e	5
f	6
g	7
h	8
i	9
j	10
k	11
...	

text: a b c d e f g h i j k

pattern: c d e f

$$h(cdef) = h(3, 4, 5, 6) = 18 \checkmark$$

$$h(abcd) = h(1, 2, 3, 4) = 10$$

$$h(bcde) = h(abcd) - 1 + 5 = 10 - 1 + 5 = 14$$

$$h(cdef) = h(bcde) - 2 + 6 = 14 - 2 + 6 = 18 \checkmark$$

a	1
b	2
c	3
d	4
e	5
f	6
g	7
h	8
i	9
j	10
k	11
...	

text: a b c d e f g h i j k

multiplier: 10^{n-1} where n is the character position from the right

$$\begin{aligned}h(abcd) &= h(1, 2, 3, 4) \\&= 1*10^3 + 2*10^2 + 3*10^1 + 4*10^0 \\&= 1000 + 200 + 30 + 4 = 1234\end{aligned}$$

$$\begin{aligned}h(bcde) &= (h(abcd) - 1*10^3) * 10 + 5 \\&= (1234 - 1000) * 10 + 5 = 2345\end{aligned}$$

$$\begin{aligned}h(cdef) &= (h(bcde) - 2*10^3) * 10 + 6 \\&= (2345 - 2000) * 10 + 6 = 3456\end{aligned}$$

a	1
b	2
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k	11
...	

text: a b c d e f g h i j k

$$\begin{aligned}h(\text{abcd}) &= h(1, 2, 3, 4) \\&= (1*10^3 + 2*10^2 + 3*10^1 + 4*10^0) \bmod 113 \\&= (1000 + 200 + 30 + 4) \bmod 113 \\&= (1234) \bmod 113 = 104\end{aligned}$$

text: a**b c d e**f g h i j k

$$\begin{aligned}h(abcd) &= (1*10^3 + 2*10^2 + 3*10^1 + 4*10^0) \bmod 113 \\ &= (1234) \bmod 113 = 104\end{aligned}$$

$$\begin{aligned}h(bcde) &= (((h(abcd) - 1*10^3 \bmod 113)*10) \bmod 113 + 5) \bmod 113 \\ &= (((104 - 96)*10) \bmod 113 + 5) \bmod 113 \\ &= (80 \bmod 113 + 5) \bmod 113 = 85\end{aligned}$$

$$\text{Sanity check: } (2000 + 300 + 40 + 5) \bmod 113 = 85$$

$$10^{99} \bmod 113 = ?$$

$$(A * B) \bmod C = ((A \bmod C) * (B \bmod C)) \bmod C$$

$$10^2 \bmod 113 = ((10 \bmod 113) * \boxed{10 \bmod 113}) \bmod 113$$

small number

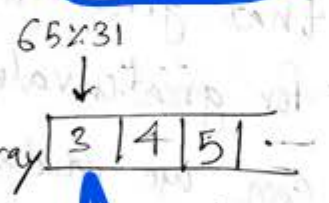
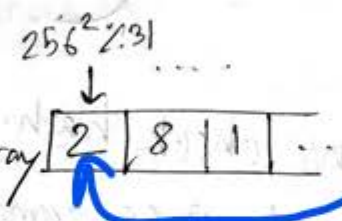
Q.

b = 256
A = 65
B = 66
C = 67

base
value to mod
q = 31

$$\sum b^{m-1} * t[i] = \text{hash function}$$

we calculate mod and store values in array for reuse



we calculate mod for the given value as well

then we use values and put in the hash putting given values

$$65 \times 256^2 + 66 \times 256 + 67$$

$$3 \times 2 + 4 \times 8 + 5$$

$$(((h(abcd) - 1 \times 10^3 \bmod 113) \times 10) \bmod 113 + 5) \bmod 113$$

$$(((104 - 96) \times 10) \bmod 113 + 5) \bmod 113$$

$$(80 \bmod 113 + 5) \bmod 113 = 85$$

256³ ← not needed

256² 256¹ 256⁰
A B C

A B C A
we are concerned with the search range

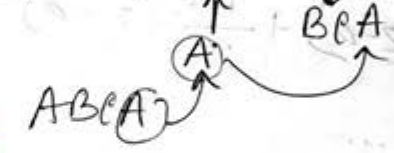
finding BCA

$$\Rightarrow 12 - 3 \times 2 = 6$$

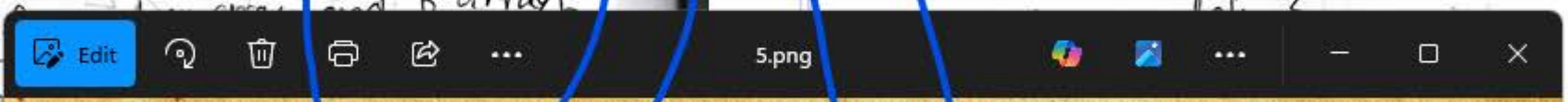
$$\Rightarrow (6 \times 256) \% 31 = 1536$$

$$\Rightarrow 1536 \% 31 = 17$$

$$\Rightarrow 17 + 3 = 20$$



Rehashing and complexity



256^3

← not needed

256^2 256^1 256^0
A B C

266

256^2 256^1 256^0
A B C A

we are concerned with the search range

finding BCA

ABC

A

BC

$$= 6$$

$$\Rightarrow 12 - 3 * 2 = 6$$

$$\Rightarrow (6 * 256) \% 31 = 1536$$

$$\Rightarrow 1536 \% 31 = 17$$

$$\Rightarrow 17 + 3 = 20$$

A

BCA

ABCA

$$(12 - 3 * 2) * 256 + 3$$



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$$(((h(abcd) - 1 * 10^3 \bmod 113) * 10) \bmod 113 + 5) \bmod 113$$

$$(((104 - 96) * 10) \bmod 113 + 5) \bmod 113$$

$$(80 \bmod 113 + 5) \bmod 113 = 85$$

256^3

← not needed

256^2 256^1 256^0
ABC

266

256^2 256^1 256^0
ABCA

we are concerned with the search range

finding BCA

ABC

A

BC

$\Rightarrow 12 - 3 * 2 = 6$

$\Rightarrow (6 * 256) \% 31 = 1536$

$\Rightarrow 1536 \% 31 = 17$

$\Rightarrow 17 + 3 = 20$

A

BCA

ABCA

$(12 - 3 * 2) * 256 + 3$

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$((h(abcd) - 1 * 10^3 \bmod 113) * 10) \bmod 113 + 5) \bmod 113$
 $((104 - 96) * 10) \bmod 113 + 5) \bmod 113$
 $(80 \bmod 113 + 5) \bmod 113 = 85$

256^3

← not needed

256^2 256^1 256^0
A B C

266

256^2 256^1 256^0
A B C A

we are concerned with the search range

finding BCA

A B C A

A

BC

$\Rightarrow 12 - 3 * 2 = 6$

$\Rightarrow (6 * 256) \% 31 = 1536$

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$\Rightarrow 17 + 3 = 20$

A

BCA

A B C A

$(12 - 3 * 2) * 256 + 3$

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$(12 - 3 * 2) * 256 + 3$

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← not needed

256^2 256^1 256^0
A B C

266

256^2 256^1 256^0
A B C A

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A B C A

$(12 - 3 * 2) * 256 + 3$

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$((h(abcd) - 1 * 10^3 \bmod 113) * 10) \bmod 113 + 5) \bmod 113$

$((104 - 96) * 10) \bmod 113 + 5) \bmod 113$

$(80 \bmod 113 + 5) \bmod 113 = 85$